ActInf Livestream #053.1 ~ "Snakes and Ladders in Paleoanthropology" &

Livestream | & | 2:06:31

Hello and welcome. It's March 23rd, 2023, and we're here in ActInf livestream 53.1.
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We're here in ActInfstream 53.1, and we are discussing a dyad of papers.
We are discussing Snakes and Ladders in Paleoanthropology by Hector, Carl, and Michael.
Welcome, Carl. And we're discussing to copy or not to copy with Hector and Michael.
Previously in 53.0, Dean and I provided some background and context on these papers, and today we're going to be able to share a video of the work that we have. So, we're going to be going to be able to share a video of the work that we have in the future. So, we're going to be able to feature a presentation by Michael Walker, and following the presentation, we'll have a discussion and see where it goes. So, with no further ado, we will all say hello, the non-authors first, then the authors, concluding with Michael and leading into the presentation. So, I'm Daniel. I'm a researcher in California, California, and it was a great fun and learning experience to work on the .0. So, I'm really excited to continue in the .1 today. To Dean.
Thanks, Daniel. I'm Dean. I'm here in Calgary. I guess the only thing I want to just highlight today is the fact that it's a 3-2-3-2-3, and I love that bouncing back and forth between those those two quantities. So, I'm really excited, and I will pass this to Hector.
Okay. Hello. I'm Hector Manrique, and I will just summarize what brought me here. Basically, I graduated in psychology. That was in Spain, and then I did my PhD in psychobiology. So, I came to learn the biological basis of behavior. Then I recycled into comparative psychology when I joined the Max Planck Institute for Evolutionary Anthropology in Leipzig. I was lucky that I could work with Josef Kaj and his supervision. And then I had the opportunity to work with the four species of grade eights. And after that, I came to know Michael in a conference on human evolution, and we have been working together ever since. So, we are now good friends and also colleagues. And that's it, basically. Thank you, Hector. And Carl, if you'd like to say hello?
Yes, I do. Please. So, my name is Carl Fiston. I'm a professor of neuroscience at University College London, and a colleague of our main speakers today, who I've never seen in person. But now, at least I have seen you virtually. So, very nice to meet you.
Great. Thank you, Carl. Michael, the floor is yours.
Hi, Michael. I'm very pleased to say hello to Carl, whom I've not yet met personally, but we're now meeting visually, virtually. I'm a retired professor at the University of Mulfi in Spain. I'm getting rather ancient. I shall be 82 years out tomorrow. But anyway, that doesn't stop me from

digging up Neanderthal skeletons and even earlier things here in Spain. I'm very pleased to be presenting this. It's a case of the dog that didn't bark in the night. So, whenever you want me to begin, I can begin, Daniel. Yes, the slide is up and it is all you.

Well, this is really a paper which we have out on SCI Archive in the version, in an earlier version. It's now been modified for a different journal to the one that was originally intended. But the presentation is in order to make a complicated subject fairly simple to comprehend.

I call it Snakes and Ladders because stuff appears, disappears and reappears throughout human evolution over the past at least two million years, if not before. And the question is, why is there not a continuous trajectory? Let's move on to the next one, Daniel. Right. The traditional view is the received wisdom is looking at what paleoanthropology can tell us from the outside. And there are two basic methodologies, one based on geology and Darwinism, which gives us one view of what it is all about. And another view down on the bottom right is all based on what modern people do and what modern apes do. And it's all by so-called analogy. Let's move on to the next one.

So both of these have drawbacks. I won't go into this in detail. You're welcome to take these slides away, provided you don't publish them because they do include some material drawn from other sources, which I haven't acknowledged formally. But the drawbacks to Darwinism is really the difficulty of being able to prove anything in the past in its own terms. Okay, we can accept it as being a valid way of explaining natural phenomena and evolution. The drawback to using modern analogies is the spatiotemporal incommensurability with regard to the regular irregularities and irregular irregularities of which things disappear and reappear throughout the Pleistocene record and, indeed, prehistory in general. Let's move on to the next one. So instead of looking at it from the outside, let's look at it from the inside. And this has an advantage of commensurability and compatibility at the same time with the gradual evolution of and within species by natural selection and at the same time with the documented behavior of Argenus Homo. Well, this is from Karl's and his colleagues' latest book. I'm sure you all know it. So let's move on to the next one.

Right. In the 1960s, some archaeologists were certainly reflecting, including some Cambridge friends of mine, the ostensibly, on the ostensive presence of irregular irregularities and regular irregularities in the archaeological record.

And a well-known one is the case of the wheel. Wheels were used for traction in Central Asia and the Middle East from about 5,000 years ago. They were invented separately in Mesoamerica just over 2,000 years ago. They weren't used for any kind of practical purpose, but just for moving effigies, statues and toys around. And this seems to have been completely independent inventions.

Okay. So let's say that surprising enactment with stuff was the exception in early humans.

Let's see if I can... I'm getting a little bit tied up with two screens here because I'm working on my own screen at the same time.

Cognitive surprises were probably disregarded in general because surprising spontaneous enactment with stuff

distracted attention away from daily routines.
Cognitive....
certainly occurred innumerable times without gaining any footholds as
technological or cultural if you like communal behavioural traditions passed
down for generation and innovation because this is unlikely if we take a
period let's say from two million years ago even one even to half a million years
ago that's one and a half million years that's 60,000 generations if you consider
one generation as lasting 25 years it's most unlikely that things could be
handed down from father to son over 60,000 generations so let's take a
different take on this let's go back to look at it from the inside from
considerations of the free energy principle and active influence it follows that
active influence underpins biological behaviour that has exploratory or
epistemic and exploitative or pragmatic aspects sensitive respectively to
ambiguity and risk enabling epistemic and ambiguity resolving responses to
support pragmatic reward seeking ones and therefore it is not implausible to the
snakes and ladders appearance disappearance and reappearance of stuff over let's say
the past two million years of existence of Arganus oh to no more than the extreme
rarity with which unorthodox epistemic exploration of possibly advantageous
opportunities in the human equation resulted from incipient albeit in
perceptibly evolving neurobiological propensities in brains or hierarchically
mechanistic minds to generate exploitative pragmatic attention via active influence to
cognitive surprises from untoward behavioural incidents which won't express more on to the next one so let's forget to see the next one
so let's forget about let's move on to the next one Daniel let's forget about this
uh how do i get from here to there how do i get from a lump of rock to let's say a hand axe
by spontaneous enactment what could provoke this okay let's move on next slide
i excavated the site i am excavating the site almost a million years old which has one hand axe
remains of burning lots of small stone tools as you can see some burnt ones on the bottom left but then
this is the dog that didn't bark why are there no more hand axe why do we only have one big large
bifacially shaped first cutting tool the hand axe and hundreds of tiny little tools people don't ask that
question let's move on to the next one they explain it away
boom people really didn't bother with
cutting up big animals they didn't need big cutting tools
well that's one way of looking at it prove it
people lost stuff and so there should be more of these around but there's only one well prove that
there's only one because people took them away and use them elsewhere move that the problem is these
these so-called interpretations only explain stuff away could well have been a valentine gift or not next
slide so let's get back to snakes and ladders we have fire at the site there aren't too many sites with fire

either and again uh the general assumption is that there was continuity uh once

trouble with moving my other um my other screen just a moment how can i move this on excuse me a moment

uh my eyesight isn't all that good these days no that's not doing it oh dear excuse me one minute um

um so we have the file as i just mentioned and

i don't believe that suddenly fire was brought to humans or invented and then passed on from father to

son nor do i believe as we'll see in the next slide that this was done along with an expansion

that can we have the next one daniel please um of these early homo erectus moving from africa outwards

carrying a hand axe carrying little embers and spreading the good news no next slide don't think

it works like that stuff couldn't have been passed on over all that period of time over all that space this

is not commensurable with modern human or modern ape behavior let's move on to the next one daniel so how do we

begin we begin we begin by bashing stone enactment with stuff and that started over three million years ago we know that

next one please

and then we have another problem which is that there is almost exponential increase in stuff

artifacts made by people and yet the brain as we know from fossil evidence does not follow that trajectory

which is the red one the brain increases and gradually flattens out at the same time as the number

of stuff that people make increases or almost exponentially let's have the next one so here we

have another problem to try and solve

and one possibility which was pointed out by the late professor uh philip tobias uh is that as the

brain increases we see in the paleo paleo anthropological record an increase in the uh variation in the volume of

the brain suggesting perhaps that there is some relaxation of selection pressure as homo uh evolved and we

have the next one again that's a point which is worth bearing in mind so let's think about another way of

how all this may or may not have happened

a methodological perspective grounded in fundamental biological and physical relationships

gives us a parsimonious prosaic deflationary account all these appearances disappearances and reappearances

of behavioral outcomes and so-called skills now professor nicholas teambergen who was a professor of

zoology at oxford and got a nobel prize uh was possibly fascinated with four questions let's have the next

slide and see what those are you want to know how we can relate evolution at

uh four different levels that of ontogeny and that of the mechanisms the causation

that of phylogeny and that of adaptation or survival value and

these are related uh diachronically and synchronically as one can see from this little diagram

uh into also what we might call approximate and ultimate explanatory um uh approaches

if we look at the problem in hand that's the next slide down um we can see that here we have in the

diachronic aspects paleolithic behavior in general in the synchronic one stuff appearing at different times

in the top right as far as ontogeny is concerned we have uh something which i'll mention later on but

we need to make a big deal of it that we know that there was a lack of adolescent growth splurt and a relatively
small brain in homo erectus one and a half million years ago compared with ours about two-thirds of
the side of ours when we think of mechanisms here we come to look at it from the inside we have the
matter of awareness involving active influence the free energy principle and variation of free energy
and we have what happens at the neurophysiological level uh allowing a complexity of behavior between hand
and stuff and and the stone in this case or stones we have to involve matters such as working memory
long-term memory perspective memory and indeed um constructive memory so how does all this tie into
phylogen here we have a snakes and ladder model uh based on the empirical evidence from the past
and we have survival value that clearly existed because homo erectus eventually gave rise to us next slide
please so let's get back to snakes and ladders here we have a man looking at the lock and saying how the
hell do i get this hand back and now the next one and in the same way with the next slide we have the
snakes and ladders a game of pure chance and random outcomes but but but next slide please we are not
dealing with completely random outcomes there's a big difference between on the one hand the
spatio-temporal snakes and ladders game of chance as if throws of a dice alone thought about appearances
disappearances and reappearances in behavioral phenomena unrelated to responses of the hierarchically
mechanistic minds of early homo and on the other hand a snakes and that is analogy for regular irregularities
and the irregular irregularities in the behavioral record owing to individually do syncretic or an
orthodox activity and the receptivity to it or not of an observing hierarchically mechanistic mind
whether that's an onlooker a bystander or that even the south same man making the or woman making the
hand acts in this case okay let's move on to the next one and here we begin to see that the heterogeneous
composition of a picture of appearances by chance alone disappearances by chance alone reappearances by
chance alone then is it useless as a model for making predictions that can be compared or contrasted with
archaeological findings of early parallel science whereas our preferred model by focusing on the
information theoretic and energetic constraints on biological neurobiological and psychosocial
evolution affords us a picture that lets us consider whether the model's focus could offer a rational
interpretation of the material record and in particular of the uneven distribution across the old world
things like stone hand axes and evidence of the tending of fire or even visiting caves things that
chimpanzees and dwellers don't do between two million and let's say two hundred thousand years in this case
eighty thousand generations okay let's move on very quickly and the rest of it in fact you're welcome
to uh read for yourselves rather than have me read it out to you because basically um you can take this
away or copy it or download it what we are arguing is that by about a million years ago adaptive evolution
was underway of cerebral human neurobiological propensities that support prospective and constructive
memory in making artifacts or working with stuff doing unusual unorthodox things together with short-term
working memory including that of the hands haptic working memory long-term working memory and long-term

procedural and episodic memory those propensities were phylogenetic outcomes of existential ontogenetic
programs adapted through natural selection for satisfying the bio energetic requirements of organisms
our genus in particular so here we have the basis of our argument that in a small band of communicatively
challenged homo erectus one member spontaneous enactment often was went district unregarded disregarded
it was regarded as useless pointless and unhelpful to gathering the food requirements needed for
surviving from day to day and we can refer the matter to the free energy principle applied to
ecological niche construction variation on your orthology to use as a fissons phrase whereby selection
of adaptive actions of organisms involves their assessment of the alternatives in terms of the
expected variation of free energy expresses the combination of epistemic and pragmatic affordances
so these early individuals a million and a half years ago who experienced cognitive surprises that's to
say prediction errors about the effective uh variation of the energy were unlikely to introduce novel
behaviors whether affected or seen by them that could gain a foothold in their community before
those communicative skills had developed in some groups that widened their social scope for active
influence and involved by natural selection within a shared framework of adaptive prior beliefs favoring
alignment of common expectations and joint activities that affected their ecological niche well
that's the basis of our argument um uh of course it's no more uh as it were uh falsifiable or
refutable within the terms of the data from the past than the alternatives that i began with but at least
it makes rational coherent sense and fits in with the uh darwinist view of evolution of species and within
species by natural selection it is a more prosaic and deflationary argument than the ones that people
archaeologists and anthropologists like to use uh and which i began with so you can take this away and read
it carefully uh what we're arguing is about a million years ago uh it's all right can you move on daniel
and again so let's just uh stick with this um well i think we can i think we can go through this
further quickly move on to the next one uh if it would yes sir right let's go let's go back to a million and a
half years ago let's look at the stuff we know there's a skeleton of a kid only nine years old according to uh
uh professor christopher dean of university college and uh that's a professor of dental anatomy and he studied the teeth
and whilst at one time uh this individual had the same height as myself i'm about um um i'm older short i'm about one meter uh 57
um nevertheless although he had our adult size he only he was only a child at nine years old and his
skull had almost certainly stopped growing it was only two thirds or less than the size of ours but he
was these people were clearly making hand axes and lighting fires or at least carrying fire around it from
time to time and visiting caves you know there was one in south africa pond of that cave and doing stuff
that uh is sort of halfway between what a chimp does and what our modern primitive peoples are
supposed to have been able to do let's say 10 000 or 20 000 years ago during the last night age
but there's a big time difference here and when we think about this kid clearly he was developing much
sooner than we do he became an adult probably before the age of 15 years old whereas our brains are not
fully formed for at least another 10 years so this would have limited let's have the next slide um uh

you can take these away and look at them i don't i don't have to read all this out in detail to you
but in uh in short um these people were precociously developed and therefore much more limited they'll
be behaving perhaps more like let's say um a modern i don't know four or five year old child of modern
adult height and with a small brain well they were not doing too badly if they kept uh uh making
uh things making stuff uh with their hands doing things and then forgetting about them because their
main purpose was to survive and they had to eat and find something to eat and they couldn't spend all
their time in these rather unproductive activities so let's say that um let's take the um active influence
approach to this and suggest that from the perspective of the free energy principle and active influence the
differences uh that we're seeing uh contextualize the experience dependent learning and planning and
influence on a small scale but they reflect the structural neurobiological priors
that could be inherited by epigenetic and genetic processes within the hierarchically
uh mechanistic minds that were evolving and which certainly were not the same as those of modern chimpanzees
and uh gorillas that don't do all these things that even our homo ancestors were doing one and a half
million years ago well okay i accept that the uh absence of uh evidence need not imply uh the evidence of
their absence but that fails to satisfy most people or many people particularly scientists like ourselves
certainly uh the darwin's argument had power has power but if we take a look at early human evolution
from the perspective of the free energy principle our proposal invokes an overarching methodological
paradigm grounded grounded in an existential principle fundamental to the evolution of living organisms
and is consistent both with the long periods of time and the vast periods of space in the old world that
we are dealing with let's say 60 000 or even 80 000 generations across three continents and
and nevertheless uh explain allows us to interpret from the point of view of natural selection uh what was
going on albeit imperceptibly slowly by natural selection and adaptation
at least some of the irregular regularities regular irregularities of phenomena attributable to homo
in the pleistocene record reflect the kinds of plausibly anomalous behavioral outcomes that doubtless occurred
often in a snakes and ladders model of evolution in early members of our genus regardless of what species
we want to call their skeletal remains whether neanderthals homo erectus modern homo whatever
okay these are plausible uh outcomes of behavior by individuals who fail to maintain orthodox behavioral responses
when faced with cognitive surprises and who showed almost certainly particular neurobiological
propensity for exploring unorthodox possibilities a propensity that's involved further in humans and in great apes
which are leak and like even our very young children seem unable to envisage how things might be as well as how they
actually are as the canadian psychologist and climatologist professor and rusten uh has written and with that i will uh end my little uh introduction
which i hope will um explain a rather complicated approach to producing a novel interpretation of uh uh how uh human behavior

was evolving during a very long period of time and across a very large period of space on our planet during the pleistocene thank you

thank you michael yes that is the end of the presentation

great

thank you for the presentation we will now move into discussion so i'm gonna

open up a blank page and then hector carl and dean please feel free to join

dean first i want to just first of all michael i don't know i'm not sure what your stature is in person but

after that presentation you're a giant in my mind so and that's that's not said out of

out of me needing to compliment you i think that thank you for the presentation i'm curious now

because the approach that daniel and i took was one of trying to parameterize a space between the two

papers i'm curious now what hector thinks in terms of sort of holding up that other that other side of

this set in particular with the paper on to copy or not to copy and how we might now sort of gain some

depth perception around things like zone of abounded surprise now that we've had sort of the introduction

with the presentation that michael just shared

hector feel free to give a first comment wow okay so you want me to comment on to copy or not to copy

on this one or how they yeah there's all about it surprise them in particular given sort of the

groundwork that michael has just established okay perfect i would like to start by thanking both

michael and carl for uh i mean this collaboration because i think i was thinking in the aphorism by

newton well attribute to newton is not original from him of on the shoulder of giants and and i think i

i feel a bit shy here among this huge intellect and and i feel very privileged too so i want to thank them both

and carl especially because he has been so kind as to revise our two papers i mean in one he's co-authored

but in the other he revised it and give us very good indications and advice so i'm very grateful because we all know how busy he is so he took the time to help us and i'm very grateful

and coming back to the the papers the thing is that michael came up with this original idea of explaining this

appearance and disappearance of hand access in terms of the free energy principle um and when we were

working on that paper i thought that the same argument could apply naturally to the case of gray dapes

and the thing that there was this puzzle in great days um studies when they studied this kind of social

transmissional culture or cumulative culture or cumulative technology and what happened is that

there are many reports on chimpanzees being innovative meaning innovative like they invent new things new

behaviors or new exploitation of resources by applying new tools so there are many instances of this

technologies but there is a puzzle here as there is the other puzzle with appearance and disappearances

of technology that there is no cumulative technology so this is a puzzle for me because if you have one

creature who is able to be so inventive how is it possible that it doesn't accumulate complexity in the

technology so the same cognitive machinery so to speak should be able to produce more complex artifacts

so there is no kind of accumulation of complexity across generations and that's a big puzzle for me so

i thought maybe the free energy principle was useful here too so i just made this analogy so it's michael

who opened up this new avenue thanks to the work of of karl and i just took advantage to think maybe it is

because new things are too surprising for gray waves so there are features that cannot cope with

high levels of surprise and i mean if i'm talking too fast or too much just tell me i think you're
following the argument i assume is it okay so the idea is that if something that i observe is too
surprising i have to deal with this high level of surprise and also by reading a paper by andy clark
was the title was relative coding it was in behavior and brain sciences if i remember correctly
he talked of one can see very surprising things if he's able to credit the sensory information as opposed to the
top-down prediction and this is the point of view of the observer of an innovation if something is too
surprising i have to consider if i trust the sensory information as opposed to the the top-down prediction
and this kind of i need to decide i mean i'm talking and i need to decide like it was some kind of
animistic thing but you you know what i mean so this is the wake waiting up of sensory information
and a top-down prediction so my idea was that perhaps what happens here is that this
gray waves when they observe something new and it's surprising they kind of do not credit it so it's like
they basically do not register that because it's um kind of is a mismatch with the prediction
and for some reason they are not able to credit this central information and this brings us to the
concept of the zones of bounded reprisal so this zone is a an imaginary concept is something we we used to
explain what happens so it's like if you have a wide zone of bounded reprisal it means that
things that are not in accordance with your top-down predictions things that deviate much from these
top-down predictions can be can be handled because you are able to bounce reprisal so you have cognitive
mechanisms that help you to bounce reprisal even when you are facing huge mismatches
with regard to your expectations to previous expectations so that's will be
the main thing here so we propose that in gray days there has not been these changes in the brain
that allow them to bounce reprisal where things deviate markedly from when what they expect
or indeed when they deviate slightly from what they expect so they can accumulate complexity provided the
things they observe that are new are not um very different from what they expected and i think i
will stop here even i can make more precision but that is the basic idea i don't know if michael is in
agreement or he wants to add something because i hope i have my limitation with language too so perhaps he can help me out
that's not really but fundamentally i accept what hector is saying his idea
was certainly novel to me when he put it forward that we have to think about the observers not merely
the practitioners because anthropologists and archaeologists and indeed many people working in the
cognitive and psychological sciences have tended to put the emphasis uh in terms of skills on the person
who performs them and not the person who uh or not the observation of those skills which is so necessary
of course and i didn't realize this until hector pointed it out to me uh so necessary of course
for their transmission to take hold at all and this struck me as being a most important insight
uh so obvious that one wonders why one never thought of it oneself in the same way
that one wonders why it is that so many paleoanthropologists and archaeologists uh are more concerned
with the stuff than with why the stuff isn't there um these are such obvious problems they go back to
sherlock holmes and sir arthur conan doyle um but we tend to be obsessed with what we can see and feel and

touch rather than um with uh what sort of process might have um uh led to explaining both presences and absences at the same time and this is where uh i find the whole notion of uh active influence uh so important to uh tie in with uh evolution um it was professor christian himself who um enthused me to go back to the lectures that i used to attend 50 years ago when i was an undergraduate uh that were given by uh nicholas tinbergen of oxford uh i would steal away from the lectures of anatomy given by sir wilfrid the grow clock to attend those of nicholas tinbergen and other people and uh i've come to realize just how important uh lawrence's and tinbergen's and um of course van fish uh and von fish as um daniel knows uh were and how worthy they were of getting 50 years ago nobel prize for physiology or medicine um um 50 years later we're still trying to get to terms with the significance of their insights which shows just how how difficult it is for us to absorb novelty but nevertheless um uh i'm no giant i'm a dwarf we stand on the shoulders of giants not only lawrence and tinbergen fish darwin of course and um going back to bayes in the 18th century one critic of our um sciarchive paper said that we were that we were um pomo in other words we were uh pomo post-modernist no no we go back in a good scientific empirical tradition and theoretical tradition back to the 18th century nothing post-modern about us at all i think all of us would reject any suspicion that we were uh involved in post-processual and post-modernist argument uh unlike many of our uh colleagues in what i call the social studies i like the late lou binford an archaeologist of america some note i prefer social studies to social science which i think can at times is something of an oxymoron um i suspect that um others will dispute that with me however um i'm no i'm no giant i'm a dwarf it took me all my time to cope with something we were never taught at school matrix algebra which i had to learn when i was doing my research on um prehistoric skulls uh and from then to cope with bayes theorem and the approaches or the uses that can be made of bayes mathematical statistics uh using um using uh matrix theory uh uh which i've gradually uh caught up with uh and i can follow it but i certainly wouldn't be capable unlike professor friston of uh yeah of um deriving it um but i can certainly understand the significance uh of something that i think a lot of our colleagues in the humanities and social studies uh simply are not prepared to uh confront at all i think they prefer much more subjective if you like post-modernist uh thinking to rational uh scientific analysis well thank you
carl lost lost the screen can you see me i can see you i can see oh we we see you carl to um um so i tend to be my eyesight is no longer quite as good as it used to be uh so i sometimes tend to huddle over the computer it's all good it's all good carl maybe you could come in and um share how this collaboration came to be or from your view from the inside or however else you'd like to comment on what we've seen so far yes well thank you very much for the opportunity um i i i also have to do a lot of huddling because of my failing eyesight. So just to reiterate, what a joy it was to hear that presentation and Hector's follow-up. So this is, for me, an adventure well outside my comfort zone. So I'm here as a passenger spectator and learning a lot. I didn't know half of this stuff, even though I did a lot of biology and a little bit of evolutionary biology as a young man. But this is intriguing and foundational insights into the way that we are.
So I'm going to make a few comments. It's going to be more of a stream of consciousness, but I qualify

this again. This is completely outside my comfort zone. I'm just talking as a lay person here or a lay physicist, perhaps. And I'll just make a few observations. But I mean, Daniel, I've been particularly interested in your synthesis because, of course, you have a, probably more than any of us, the broadest oversight from the evolutionary perspective right through to the Bayesian mechanics that is entailed by things like active inference. Just to forewarn you, I'm going to call upon you to do the synthesis and to draw and join the dots that haven't already been joined, although there's been a lot of top joining from what I can see already. So the stream of consciousness comes really from this

commitment to ideas that are certainly not postmodern. I think I'd probably have to go back to Plato and certainly sort of get stuck around the ear of Helmholtz and the like, you know, from the from a physicist's point of view. And pursuing that, looking at the looking at the this sort of remarkable snakes and ladders phenomenon from through the eyes of a physicist. And it certainly makes complete sense that if we're looking at a process that would be described as a non-equilibrium steady state that is slowly evolving, that could be, if you like, a mathematical image of the evolutionary process. The definitive feature of these non-equilibrium steady states that have

the biological aspect is that they entail a revisiting of the trajectory to neighborhoods of certain parts of phase space.

So if I read phase space as basically the state of evolution or what what phenotypes are doing or what is encoded genetically in a very, very crude, broad level,

there can be no other solution to a particular trajectory that is apt for describing a non-equilibrium steady state than the occasional

a inherent recurrence of certain states of being, literally being close or revisiting neighborhoods of certain regimes of phase space.

So the the snakes and ladders image seems to me a very apt image to actually describe

pullback attractors, which are another way of articulating these kinds of dynamics or this kind of

kind of dynamics in the sense that they have this aspect of recurrence and they have this aspect of self-contained itinerancy,

which are the hallmarks of any system really that exists over a substantial period of time.

It would be interesting to think, can it be any other way? And I would submit no.

You know, if you have a physical process that is unfolding at multiple timescales and it is endowed

and can be recognized either historically or in the moment by being in certain characteristic states that literally

characterize the system in question so that it can be named, whether it's a species or whether it's your behaviour or whether it's an artifact or whether it's a process, then it must be the case that it has this it has this kind of utterancy where things come and go. So it will be interesting just to you in a as a pure exercise for pure mathematics to support things like the Boncari recurrence theorem into a snakes and ladders format and vice versa because I think there's something fundamentally true that has to be the way that things are. And I see that in many fields and two immediately came to mind.

The first was current notions of the, not the spread of ideas or the spread of tools, but the spread of cancer

you know, the view that cancer, there are two views of cancer. There's a sort of 20th century view that you get
one cell in your body fails to recognize that it should not be growing and just keeps on growing and
you'll get these progressive and inevitable growth of the cancer set of cells and at some point they will
disseminate and seed themselves around the body. You get the pastases and secondaries
and the like and ultimately unless it's cured then it will it will destroy the phenotype. There's another
view which is is now becoming more prevalent which is a much more snakes and ladders view
and this view is that we're all the time generating little cancer cells but 99.9% of the time they don't
get a foothold. So that we're in equilibrium, hopefully on the right side of an equilibrium,
always playing host to little cancer cells. It's just that it's very rarely do they actually establish
themselves to the extent that they can get a foothold and reproduce to actually produce a
tumour or a neoplastic growth. And that struck me, it would be nice to put a snakes and ladders
sort of picture on the way that that particular biological natural system is viewed and it's
certainly very consistent with current views on cancer. The other thing, the other system
that seems to comply with this notion is not again the spread of ideas or the spread of tools or cultural
means but the spread of viruses in epidemiology. It's very interesting that there are certain
viruses for example that get a foothold and explode exponentially. We've had you know we've just enjoyed
or witnessed an episode of that with the with the recent Covid-19 pandemic but there are other
very very very similar episodes that don't get a foothold and they die out very very quickly because
they can't spread. And I know this personally because we were looking for data when modelling using
actually exactly the same maths that underwrites the free energy principle active influence. We were modelling
the coronavirus prices recently and we hope to get some priors on certain parameters by looking at
legacy historical data and it's interesting that there is data out there on very similar
viral epidemics but they died out too quickly to generate enough data to make sense of. So things
like MERS for example was certainly recorded but it went away before it actually produced enough data to get
a handle on it. So again you know this phenomena of sort of sight things repeating themselves you know
you know climbing up the ladder but then sometimes falling down a snake very very quickly so that
they you have to wait until the next occurrence seems to be a fundamental fundamental aspect of
ensemble dynamics wherever you see it either within the body in terms of neoplasia or in terms of
epidemiology and the spread and I use the word spread deliberately because of the you know that
that latter perspective on basically to copy or not to copy. So if you're copying then you then you're in
the game of spreading and if you don't copy then you don't spread. So I think just coming back to you know
the question you know what what what is it that determines whether a particular cognitive surprise or
a novel approach to you know a way of behaving or indeed a way of being catches on or not. And you know that
dialectic I think is you know it's absolutely intriguing. I should just say that of course

the all of this formulation rests very very sensitively on a separate of temporal scales you know and you know we have to sort of be in the background there is always this separation of scales that you get with evolution.
And in this particular instance what we are talking about is really whether certain things can survive generationally in the spirit I would imagine of sort of evo-devo and cultural niche construction.
And then the question then is okay at a fast time scale of you know interactions between phenotypes and what's going to make that sufficiently endemic to be propagated at a transgenerational time scale.
I think that's a really interesting question you know again it's a kind of question which as a neuroscientist and certainly as a clinician you are often confronted with you know things for example like epilepsy you know what is it that constrain that determines whether a particular epileptic surprising dynamic discharge in a particular part of the brain can be propagated and disseminated throughout the brain to give you a fit or as opposed to it sort of auto-vituating or you know quenching itself because the neighboring cells are just not interested they just don't recognize that particular pattern and it's inhibited. And I guess the answer I mean the I think we've already had the answer both Michael and Hector have articulated it but I guess the answer lies in the mechanics of the spread of ideas and shared frames of reference and whether it is too surprising to be incorporated into your conspecifics or your community or whether it is learnable and can be incorporated or absorbed into your generative models such that it can then be spread to the next person you know in the standard sense of sort of niche construction. And Hector mentions specifically one key determinant of that which is the um perhaps it was Michael but it was the um the ability to um change your mind um and you know the ability to change your mind is absolutely fundamental if you just you know if you want to frame this in an active inference context um just confronted with a cognitive surprise as it has been articulated a set of difficult to explain um sensations or sensory evidence then there are two ways in which you can resolve that surprise the first way is to change your mind and to adjust your internal generative models or your do some basic belief updating on the inside to make sense of this thing and that might in uh depend upon some experience dependent learning uh over a suitable time scale um so that you're changing your model your generative model of the world to make it more apt to explain this novel observation or an observation of one of your conspecifics doing this kind of thing um the other thing the other way is not the um the perceptual response that the belief updating on the inside to make your predictions more like the sensations it's to actually act upon the world to um to make your sensations more like they were predicted so this is a fun this is foundational in active influence and this two complementary ways of minimizing surprise or or prediction error you either change your mind or you change the thing that you're surprised about um you're at every level at every temporal scale so just read in terms of whether i
um um i am able to change my mind about witnessing fire there are two ways i can say yes fire is a thing and you know i can then incorporate or imbibe that into my world model and um become the kind of creature that actually engages with fire or i can act to put the fire out and then i don't have to worry about it both are perfectly appropriate ways of minimizing surprise and the difference what we read at a

at a very basic level um the um the thing that determines what i um whether i perceive or act or
whether i engage in perceptual or active inference um in the moment to moment depends exactly on something
which was um again mentioned implicitly earlier on which is the precision afforded your the sensory
evidence relative to your prior beliefs so as a physiologist um if you don't know this is you know
it was a beautiful there are beautiful illustrations of this so if we if you if we just forget about the
notion um at the moment of sort of long-term revisions or belief updating of our generative model over
multiple experiences to an instance of somebody making a fire um or providing me with a valentine's
kit of a handaxe and i'm just think now to um the moment to moment ways of making uh of sampling the
world and making sense of the world um it's the case that in order to move and in order to change the
world i have to attenuate the precision afforded my sensory information because if i didn't do that
i would not be able to ignore all the evidence that i'm not currently moving so the most beautiful
example of this i think is something called saccadic suppression it's when we move our eyes and we do so
about four times a second we actually attenuate the precision of the sensory evidence so that we
are able to actually move our eyes which basically means you can't see the optic flow induced when you
make a sharp or saccadic eye movement and yet 50 milliseconds after you've made the eye movement
you're suddenly now attending to the visual information that you can see so you can't you
can't actually see anything while your eyes are moving because you've got this sensory attenuation
which is basically putting the priors um you're in charge uh suspending attention to the sensorium while you
are acting um failures of that um can have quite horrible consequences so a failure to ignore the
evidence that i'm not moving and a failure to attenuate or suspend attention attenuate the precision
of sensory evidence or afforded sensory evidence um looks like things like parkinson's disease
so you know we think that um this precision is encoded by certain neurochemical transmittance particularly
dopamine a deficit of dopamine basically means an inability to um attenuate the evidence from your
um strep receptors or all the signals coming from the body and that you're not currently moving so if
you have the prior belief i'm going to move i'm going to sit up walk um drink from this cup you may have
that prior belief but it's immediately counteracted by and dispelled by very precise evidence you're not
moving so you can never actually initiate a movement and what you actually see clinically is basically a
psychomotor poverty or a bradykinesia the point i'm making here all reduces to something very very simple
which is the precision afforded sensory evidence and if you can't get that right and notice it has to be
estimated at every time scale either in terms of saccadic suppression or on a a 10 millisecond time scale
uh right through to um one might actually argue um um an evolutionary time scale and i make that argument
because um precision is it can also be thought of as a rate constant so if if i've got some sort of um
sensory evidence and some prior beliefs um and i now view my belief updating at any level whether it's
sort of you know my neural activity encoding my beliefs at the moment or whether it's my um my neuronal
connectivity or um you know encoding what i have learned or its neurodevelopment or indeed at an evolutionary
time scale the precision can be read as a rate constant so now what we're talking about is a certain

quantity that is absolutely crucial for whether i um adopt this um or incorporate um instill this novel experience into my model or not that depends upon a precision or a rate constant which means that um one would um if one wanted to ask the question you know why do some species have the ability to um um to learn
uh whereas other species don't then i think the the answer is quite simply in their ability to um to optimize and adjust their learning rate either precision afforded the evidence at hand that drives the learning um and i i i'm wondering whether there are a couple of ways to take that that sort of um mathematical take on this on this really single really important quantity um and just to make it intuitive i'm trying to get back to the copy versus not copy so if i can copy that means that i am um i'm impressed impressionable what does that mean it means that i will accumulate new evidence very very quickly what does that mean that means i have the capacity to assign greater evidence to or precision to that sensory evidence and suspend the precision of my prior beliefs i'm not rigid in my thinking i can learn because i am impressionable um what would that look like what would look like you were young it would look like you have just been born and you are designed to learn because you you have yet to build neurodevelopmentally your model of the world um which means that um averaged over a lifetime the the degree to which you are likely to learn and implicitly enable the spread um of ideas um and you know ignite this particular um episode of um your of of a new tool or a new a new way of doing things or um um being um then it would look as if you are younger for longer so i'm just i was just intrigued i i think i think michael you know was making an explicit reference to the fact that um that it may be some there may be a key difference between certain species maybe the um the duration or the timing of the point where you basically switch off your learnability you switch off your uh impressionability and you become an adult and you bake in then and then from that point on you become old and wise and stuck in your ways and instead of actually trying to change your mind when something changes you don't you become not conservative in a political sense but you actually uh try to keep things status quo uh committed to the social norms um but before that you don't you explore you go to discos you do bungee jumping you you know make new friends um but if it's the case then that the the the spread of ideas depends upon being a child at heart for longer then then that observation about at what point do we um get to that epigenetically determined stage of neurodevelopment becomes absolutely crucial and just even longer this is the final point um an even longer time scale i'm wondering whether the same maths can be borrowed or the same concepts at least can be borrowed um to think about um this kind of evolvability um uh from the perspective of people like stuart kaufman and like so selection for selectability so again um if you're now reading these rate constants or precisions um uh in an evolutionary setting what you're what you are talking about is basically the the you know the mutation rates or the selectability um which of course itself can be subject to selective pressure so you know if you um if you commit to this style of thinking about belief updating in a in a mathematical sense at many different levels or different spatiotemporal scales or these temporal scales um then it may be that the the the selection for selectability or the selectability in and of itself is um exactly the same kind of quantity that is the you know the learnability and the you know being young at heart for longer

that just is the same kind of um quantity that is uh required right down to the level of uh
saccadic suppression and sensory attenuation that determines when we act to keep the world constant as
opposed to no we actually have to attend to this evidence and and update our beliefs um and um if
that's the case then um this notion uh well i just wanted to to close on on on sort of the importance of
um cultural niche construction um looked at um through the lens of these this um rate of um adaptation
precision selectability at different levels and how all of these couple to each other in a sort of bootstrapping
way and whether that would be your useful view of um the spread of the spread of ideas
um sometimes spreading and sometimes not spreading um can i just ask it that did we um did we or did did
you latterly um sort of leverage um sort of the well actually know this is basically setting up daniel
for his his perspective but i'm just wondering how much um people in um evo devo and uh evolutionary
psychology i'm thinking here people like cecilia hayes and uh cognitive gadgets how much they they
think about these um these issues and whether there's the opportunity or whether they have a snakes
and ladders like um view of things uh from the point of view of um evolutionary psychology and that's it
that's all i have to say thank you daniel your turn
thank you wow
barely know which side of that hand axe to flake
um the multi-scale learning is a great point and perspective we can just imagine the extreme cases
in the total acceptance of sensory evidence world then that thing defined by its revisiting
potentially infrequently of characteristic states that thing ceases to be that thing in that whether
that's seen in some positive um light a neutral entanglement light or a death-like light it will
cease to be that thing because it will have just simply informationally if not biomaterially accepted
the generative processes in positions and then the other extreme case would be no attention uh see no
evil hear no evil and in that case the trajectory from a bayesian mechanics perspective is massive it's a
uh trajectory with so much inertia that it can't be moved and that sounds like it would ever happen but
in fact when we heard from hector and michael that like upon seeing an innovation a great ape does not
take up that innovation they're prior on how to accomplish a behavioral outcome or even at a higher
order some functional outcome can be seen as massive the bayesian physical inertia of their top-down prior
dominates these potentially sparse or non-linearly related cues coming up from the bottom so simply by
not paying attention to the bottom-up cues
there is the imposition of this top-down mass and so that's a bayesian mechanical way to understand
well if we had no priors at all no prior no problem but then we dissolve but if we have um the titanic
prior then it's a fragile strategy and so if the niche were very simple and unchanging that may be locally
an adaptive strategy and on certain axes of phenotype where there can be regularities we do see those kinds
of strategies like a bacterium that has selected to live in a constant temperature geothermal niche
may come to entrench in its protein structure and in its gene regulatory networks and so on
an embodiment that sharpens and sharpens and sharpens its capacity to compete in that regularity
from a temperature perspective and therefore at the detriment of fitness in changing environments
so that's one aspect is seeing learning as this multi-scale

synergetic and ongoing renegotiation process of the updating across multiple scales
some spatial temporal scales of which there are no problems with talking about those using the
cognitive ontology like perception cognition and action but then some scales especially beyond the
physical mechanisms that those cognitive ontological terms arose from
states states states states states states states states states states states states states states states states
interoperable mathematics and statistics and analysis frameworks so we can just qualitatively
even in conversation start to see that as a starting point instead of needing to argue
up to that point of that being important for example in pursuit of transdisciplinary synthesis
conceptually epistemically or so that we could explain predict design control create
pragmatically how can we take that multi-scale insight as our starting point and then i think
that leads to a question in the chat which i will ask now mark fruchtman wrote early in the video
was the complaint that there was a lack of evidence as to why certain things occurred in the archaeological
evidence record how does active inference not suffer from the same issue and so it's a key
question thank you mark for asking and was the motivation for one of my underlying motivations
for even reading and investigating with this lens was great if we can describe everything and we have
this very interesting mathematics and so on how do we go from a technique and a method to
moving beyond some limitations of previous analysis techniques
we can go to dean or whomever else has a thought
we can i can throw something in but i'm not sure it's it's going to satisfy directly the question
um maybe somebody else wants to throw something about here i i do want to i want to indirectly answer the
question and i'll come and i'll loop into what i think might be satisfying part of it i think one of the
things that that in particular that to copy or not to copy paper emphasizes is the observer's perspective
so rather than just looking at that as a as something written down on the page taking up or absorbing that
position the observer's perspective i think what the two papers and one of the things that we tried to
highlight in and we're so grateful that michael and and carl and and hector were willing to let us look at two things at once
was the idea that the papers um both raised the specter of and i put quotes around this
the narrative follows what the implications of start with a story about seeing phenomena x actually are
now one of the implications of that is that when you have a story you can't have gaps in the plot so
people who are are really story biased and story forward they want those gaps filled in as quickly as
possible that's how we comprehend but the interesting thing is if we are actually actively inferring as
opposed to just seeing it as a subject matter or a topic we can take a game theory approach and in games
there will be random gaps like charles launders pierce says an exhaustive inquiry well if it's exhaustive
we're also going to need recovery and reflection and and so on so there will be gaps and so the snakes and
ladders metaphor to me is like okay perfect now we're starting with a game as opposed to the retelling or
the manufacturing of starting with a story so there's very different mechanisms in in the two approaches
right there are are there are very different assumptions in the two approaches and then the
question around modeling as per as perspective swapping kind of pivots on the stuff that michael was

talked about in terms of how is the change based model depth capacity either affording or not affording that ability to play with time as opposed to being captured by it because in a story you've you've written down what the sequence is it's now been applied now you can interpret that but in terms of that actual effort of starting with a story it's now stabilized per se relatively speaking so

let's start with a game and carl's heard this before because i asked him a question about a year and a half maybe two years ago is starting with a game is a prediction matter expertise strategy or a bias starting with a story is a subject matter expertise strategy or a bias

a bias and you can understand why if you're if you're dealing with historical records

to fall into a default of okay well we have to start with the story but i think what hector and michael are at least asking us to have a conversation around is well so what if we do bring this other way of viewing the circumstances taking that game strategy and putting it on the same shelf as the story strategy

so again if you if you look at the assumptions around why gaps have to be filled in as opposed to there will be gaps because the game is on something as simple as that and being able to hold the two things up at once can sometimes at least tell you a little bit about maybe some of the

some of the conditions that you as the observer are inserting on what it is that you're observing or example

was it sherlock who said the game is afoot

maybe right

yeah how the bus

okay yes carl indeed a michael or hector

how is this landing with with what you're thinking right now michael go ahead you always know better then i can pick up from you

i'm not quite sure where to begin um perhaps thinking in terms of the evolution of both the brain and the body in the genus homo uh we and possibly neanderthals also maybe we're going to be a little bit of a maybe from as early as big quarters of a million or even or at least half a million years ago

seem to have undergone an evolution towards a bimodal growth spurt or growth growth curve

caused by the adolescent growth spurt following a period period of childhood slowing down of growth growth which is unique in our genus to our genus and

seems to be related in adolescence to some changes in terms of the agility to react to complex situations or not as the case may be but even before that in early childhood

i'm now talking about the first three or four years of life our infants seem to have exceeded already quite considerably their cognitive ability particularly with regard to working memory when contrasted against the present great apes in other words there has definitely been not merely an expansion of the brain but in particular changes in the connective systems within the brain which clearly must have an evolutionary physical basis in terms of uh structure both um uh inter and in particular intran

neuronal structure which when i began to study medicine physiology over a long time ago now 1960 um we could barely imagine

uh it was only in the very end of the 1950s i think 59 if i'm not mistaken professor friston can correct me that uh at cambridge henry huxler was doing uh and i began to do inter neuronal studies on on accident potentials

uh we were doing them in the lab at oxford as undergraduates about four years later and um then

i was doing them under professor powell gleese at getting him a few years later still uh with a practical

application um because we were studying uh the chain the changes in these following orthocresal phosphate

poisoning um which had a structural situation in hens which we were working on similar to those in humans

who had suffered from a massive um outbreak of poisoning at agadir in morocco when a lot of

local people were sold american jet fuel oil rich in orthocresal phosphates for cooking with

and there was an epidemic of many deaths um which is well known in in medical history and so some of these

studies uh early studies uh even then were having applications which now uh seem almost paleolithic

when we think of the work which is being done on intraneuronal and also on synaptic connectivity

into intraneural neuronal transmission and the effects not merely

of genetics such as for example synaptic cooling but also of epigenetic modifications which no doubt

we shall learn a lot more about in the coming decades but which even now seem to be telling us

that the increase in um the kind of connectivity or um what's the word i'm really looking for

perhaps connectivity is not the best word uh within the brain is far greater uh than anything which

is possible within uh even uh growing chimpanzees let alone uh the adults or other great apes

and this must have evolved somehow uh probably in relation to the appearance

uh i suspect of the slowing down of childhood growth uh followed by uh the uniquely human uh homo uh

adolescent growth spurt which uh was must was completely absent uh one and a half million years ago

in early homo erectus uh according to um um professor christopher dean and others and which must surely

have been a response or the evolutionary change must have been a response to not merely

not merely some kind of magical physiological change but more more likely uh to

the effects of uh what one might call behavioral responses and fundamentally it takes us back to active

influence and to fundamental changes which are naturally slow uh to evolve and this is the part which is not recognized

by most of our colleagues who work with uh living organisms or living climates and for that matter

uh living uh human communities and individuals um their perspective is not that um i hope they'll forgive me for saying this

which those of us who've been brought up in the natural sciences uh naturally come to expect

they don't have the kind of cosmic time scale and cosmic spatial scale uh of things

and of their slowness and of their slowness and the imperceptibility of change which come naturally to

those of us brought up from school days in a different tradition and i think this is very important because

i think there is a big divide still between what um six years ago uh we knew as being uh named the two

cultures i think this divide still exists um as that uh critic of our paper who claimed we were post-modernists

uh clearly represents um the more uh shall we say um um intuitive relativistic uh view uh in which

there is something which uh some of them believe to want to call an ethnographical present which i think is

is a contradiction in terms otherwise it could be could have been no evolution in behavior but which
wants to compare uh instead of contrast uh as i would have it uh what goes on in a few generations let us
say in a community be it a small one albeit a large one today with what went on over many many generations
um as i said between 2 million and 200 000 years ago there would have been at least 80 000 generations
if each was 25 years long and even that is pushing it a bit given that adulthood in homo erectus
was very likely achieved before the age of 15 and reproductive success uh was probably or the next
generation was probably um only a few years away this is important because as professor christian
will not talking about cancer um our longevity um has allowed us to and of course i'm better hygiene and
medical knowledge but particularly our longevity has permitted us to survive uh in ways that were
unthinkable perhaps um in many parts of the world uh barely um well less than a hundred years ago
that wasn't merely thanks to antibiotics or even better hygiene um in but to increases in
uh standard of living uh going back perhaps perhaps um a few hundred years but let's not get into that
what i really want to say is that
next week i have to have a small um basal cell neoplasm removed from the side of my nose
which only at the age of um 80 or 81 became uh noticeable
uh indicating that some of these um uh diseases which we we now regard as being um part and parcel of
our uh humanity or the aortic stenosis which was i was diagnosed with with my first and only ever um incident
of um cardiac failure about a year ago and has been successfully treated thanks to uh uh modern catheterization
uh process processes that allow pro prosthetics to be prosthetics to be placed on heart valves by a long
catheter from the groin this kind of thing would have been unthinkable um a few years ago in fact i can
remember as an undergraduate and a medical student watching the first ever uh open heart uh operations
done by alf gunning um which was regarded in 1960 whenever it was 63 64 as being um something that might
never might never might never succeed and nowadays these operations on our heart adjuster as everyday
cancers in cardiovascular units around the world things are changing and they're changing at a
with an acceleration which we can only barely uh appreciate though even those of us who now outlived our
four score years and ten uh and are on the verge of uh of defunc defunction however um we have to
think in terms of the and this again is important of the exponential uh if you like rather crudely scale at
which things are happening and which are so influencing many people many of our colleagues in the
uh humanities and in the social studies and in indeed studies of animal behavior um including climate
behavior of course and this i think is detracting from an appreciation of the long time scale uh and the um
over which evolutionary processes have uh necessarily still take place i don't believe in goldschmidt's
uh hopeful monsters magical genes some people believe that there's a magical gene in homo sapiens
switched on our cognitive ability i think this is this is wishful thinking um indeed it's magical
thinking in my view uh however um i suppose my own training would leave me to want to say that but i'm not
there have been exceptions and one exception i should like to mention by name
is an emeritus professor of archaeology who is eminent uh in paleolithic studies at york university now

professor jeff bailey who over 40 years ago was pointing out the incommensurability of the timescales
of recent archaeology and if you like ethnography with those of the paleolithic and the pleistocene and human
evolution and this was not really well regarded at the time although it influenced me very greatly and
so there have been honorable exceptions um who have certainly um influenced me um we do as i say stand on
the shoulders of others we try to um make one or two uh small comments advances in both methodology and in
theoretical um influences and deductions and i think that the
uh we're finally coming to possibly begin to see how to bring together uh thanks to um especially
professor friston's insights and um methodological um
um um what i'm trying to say uh methodological proposals and advances uh we're beginning to see that it might
be possible to square the circle to bring together the uh evolutionary um very slow uh evolutionary uh
development of the hierarchical um um mind which characterizes argenus and
actual um processes which can underpin this undergird it um can bring together uh the
uh darwinian view which um darwinian view which um most of us espouse in the west uh with a few
exceptions which are too well known to be worth mentioning here um i'm sure daniel knows more about them than i do
um but um it's meaning to bring together
the theory of evolution of species by natural selection of and within species and the within part is
important and it's being to bring bring it together with what we can appreciate about human behavior
which is so uh different from that of the great eight and here i just want to make one final point
which is that uh in another paper which we um which hector and i were co-authors along with
um professor um right read of ucla um in um uh science and biode behavioral reviews last year we pointed out that
there has been too much in our view uh hopeful or wishful thinking about the human nature of great apes
certainly certainly their behavior is very
different from that of many other orders of mammals and not to mention other classes within
the animal kingdom however there is a big difference and i think this difference is best
um emphasized by going back to the growth curve of humans and of human behavior in relation to
our childhood in particular our infants are capable by about the age of three of putting
things of combinatorial combinatorial thinking in ways that are virtually impossible or outside the
scope of the behavior of most great apes most of the time and almost certainly in the wild and this
i think demands an explanation which is more than simply saying no the apes are really more like us
it's just that we don't recognize this no i think that there is a difference here and the time scale
uh has to be taken into consideration which it often is not and most important and this brings
up me back to my last point and the first one which is that finally we're beginning to get together thanks
to professor christians christians insights to seeing the possibility of tying together
the biological evolution and if you like um the psychological evolution certainly

uh i would say the neurobiological evolution uh which has characterized our genus and characterizes it

today um in a unique way and with that i guess my case

thank you michael

many many ways to go the dinner plate balanced on top of the stick and i think that's the cognitive moment

where the top-down constraints and opportunities of our enabling architectures which for a long time

is known to be known

versa. And when we consider information spaces, we have similar connectivity patterns. Some are

like information anatomy, and some are more like information physiology in narrative or rhetorical

spaces. That becomes part of our extended cognitive selves and leave to others in the best possible

way those who want bright lines of what is a self. There are analytic frameworks that respect

continuities and discontinuities with an emphasis on the respect. The continuities, we want them to be

differentiable and integrated and generalized coordinates of motion and the continuous time

active inference models and the path integral formulation. And then the discrete time generative

model, which is also a map, not the territory, helps us understand saltatory maneuvers over multiple

timescales. And looking back at the literature and the discourse, discussions like, well, is evolution

gradual or general and jumping? These have been some of the biggest touch points. Are things slowly

changing or fast changing? Those are where certain lines were drawn. And the methodological developments,

Bayesian graphs, Bayesian graphs, and so on and so on, have provided a instrumentalist lens that,

as a matter of fact, does integrate these frameworks methodologically. That is a claim that hardly anyone

dissents from the same thing. And then what happens with respect to our past, present, and future,

when we realize that we're after the moment of intra and among disciplinary tension,

not pushing the rock up the hill anymore, or trying to bring somebody else along, but gardening in a space

that already is how we need it to be.

Could I make a comment, Daniel?

Yeah.

It just occurred to me that when we think of the formant frequencies that affect all of the harmonics

and other aspects of our vocalizations, whether speech, chanting, song, or even whistling, perhaps,

it is unlikely or improbable that these very many variants which occur in different languages

throughout the world with very, very different characteristics,

and which almost certainly had begun to evolve quite considerably by the time that early writing appears, about 5,000 years ago. But it's still unlikely that these can be taken back into the past on a slow evolving timescale,

such that they can be related directly in a direct evolutionary line to the vocalizations, let's say, of great apes,

from a common ancestor that we have with them, six million years ago.

So if we accept this unlikelyhood, then it seems equally unlikely to me that we can talk about cultural transmission,

or technological transmission, in a slow, uniform manner, invariant manner, over the past, let's say,

two million years, from which we have information about different technical skills and technical abilities,

or at least, shall we say, technical possibilities that were exploited by our Argenus Homo.

And therefore, I would argue that it is equally likely that the unawareness of what people,

what individuals were actually enacting, the unawareness by bystanders, onlookers, or even the very protagonists

of those activities themselves, was a continuous, in continuous evolution over, let us say, 80 or at least 60,000 generations

in a continuous invariant manner. This seems unlikely, and it seems equally unlikely,

that the same is true of different linguistic, shall we say, abilities or aptitudes.

In other words, as Professor Friston has remarked in more than one paper,

it does seem that the advantages that were taken were taken not in a continuous manner, but in a discontinuous manner,

when particular communicative skills had developed in some groups, such as to be able to widen their social scope

and proactive influence, and had evolved by natural selection within a shared framework of adaptive prior beliefs,

favouring alignment of common expectations and joint activities affecting their ecological niche formation or development.

And this, it seems to me, is a very expedient way of looking at how things happened, or are likely to have happened,

whether we will ever be able to produce a theoretic in terms of a formal Bayesian proposal for this, I don't know.

I would like to think so, but I really don't know how, I don't have the competence to begin to do this myself.

But I think that it is, if it is possible, it will certainly be, to put it mildly, a challenge to those who tend to think that there is a rapid jump.

Well, a rapid jump took place, which, shall we say, a small gap, there is only a small gap that separates us from the great apes of the equatorial latitudes of the old world.

I think that that took a much longer period of time, and I don't think that one could, I don't think that to compare what modern, even, shall we say, in inverted commas, relatively simple or primitive communities do, or how they act, is commensurable with the behaviour of modern great apes.

Frankly, I would like to challenge this, and I would hope that it might be possible to produce a model which will permit the, a theoretic, which will permit the evolutionary process to be at least interpreted in an alternative fashion.

Sorry, I didn't mean to be long-winded.

I apologise.

The classical information is inscribed on the holograph.

In the classical sedimentary layers is...

Daniel, you're muted.

Sorry.

In the holograph, we have the inscription of the classical information.

And the quantum cognitive moment is the unpacking or the rotation of what is on the holograph into the information space.

And that, I think, was provocatively raised when you asked, what if this was all there ever was?

What if there wasn't a second-hand axe to find at that field site?

That's the interpretive moment.

And how we interpret it is everything.

So, in our final ten minutes of this continuous 53.1, if each of us could provide a kind of closing, just several minutes or any several thoughts on anything they'd like to add in to this session,

or any directions that they would like to seed or prompt for next week's 53.2?

I have something, Daniel, that we'd like to contribute before we close the session.

I would like to take advantage of Professor Freestone with us, because I have some questions for him that have to do with the free energy principle and the to copy and not to copy,

if you allow me to just formulate it.

And first thing I would like to do is that there were some lapses in my argumentation before.

I want to make clear that chimpanzees learn socially.

This is something we know for sure.

So, there is social learning and there is even cultures, as Professor Andre Witten in St. Andrews has proven, has demonstrated.

So, there are some cultural traditions. What is lacking is accumulation of complexity. What is called by Michael Tomasello, ratcheting up.

Okay. So, from one generation to the next, every generation builds upon what was produced in last generation.

So, you have a tool implement and you modify it, you make it more complex.

So, our technology is the result of many people accumulating their knowledge.

So, it's kind of a joined endeavor.

This, I wanted to make clear.

And what the models proposed, theoretical and other kinds of models, is that coping with fidelity is key to have this accreation of complexity.

So, what I wanted to correct is that the puzzle is that high rates of innovativeness does not relate in a creation of complex technology because there is a lack of copying with fidelity.

What is called true imitation.

For some, and for Russo and Birne, I think they call it impersonation.

So, I see someone doing something, I copy the means and the result.

So, chimpanzees are more prompted or likely to emulate, which is copying the end result, but by their own devices.

So, they can do different things to produce the same result.

And it appears that copying with fidelity is key.

And I'm cutting this long explanation.

I just wanted to correct this for the sake of purity of science.

And now my question for Professor Freeston, if you allow me.

There is something interesting that the team of Andrew Witten in St. Andrew's observed under this in our paper.

That there was one instance that I know of when there was this kind of accumulation of complexity in chimpanzees.

And the thing was possible because these chimpanzees that combine two different techniques to make something more complex.

It could be a kind of primogenious example of what we call this accreation of ratcheting up of behavior.

They did it because they had performed or enacted these behaviors before.

And this is for me is very intriguing because thinking of how in 2010 there is this paper by Freeston and other people where they talk of the generative models under the free energy principle.

And they talk of inactive inference like generative model or not, the structural representation is nothing that you enact, you entail with your whole body, so to speak.

Correct me if I'm not, Professor Freeston.

So I'm thinking that these guys, these chimpanzees that were able to have this complex tool product, they had enacted the behaviors independently before.

So I'm wondering if this enacting the behavior before allowed them to become surprised.

Because this would be, we go to the core of this idea of inactive inference.

Like if you have performed those things separately with your body, then the surprises kind of bounded.

And you can produce these more complex products.

I mean, I don't know if I was clear enough, but this is something that is intriguing to me.

Sure.

Karl, as your last thoughts, feel free to address that or anything else.

Well, yes, I'm just reflecting on that, I think, very important question.

I mean, it is certainly the case that all of the active inference formalism, both in neuroscience, but also in terms of, you know, ensemble dynamics and self-organization of collections of

things, rests upon a generative model of the consequences of my action.

Even to the extent you could argue from a theological perspective that all our perception is just in service of deciding what's the best thing to do next and literally how to move next.

Just acknowledging that only a very limited number of ways you can actually change the universe.

You can either move a striated muscle or you can secrete something.

There's not much else you can do.

And so, you know, it becomes very important in terms of your characteristic state of being and to have the right generative models that choose the right ways to move your muscles or to secrete things.

And that places embodied action or having embodiment and activism at the centre of the generative models.

And that certainly makes sense to me from the point of view of both a statistician and a machine learning enthusiast, that it would be much easier to revise or update a generative model in the sense you were intimating in terms of accommodating the next kind of behaviour.

You know, I'm not sure whether this is composing a sequence of behaviours, but certainly being able to incrementally embellish or adopt a behaviour if you had already a precursor that was inherent in the structure or could be scaffolded within your generative model.

So, I mean, the idea here is that everything that goes on the outside has to be in some structural and dynamic way recapitulated on the inside in order for you to control the outside.

And this, of course, this is at the heart of early formulations of things like the free energy principle.

In cybernetics, you know, Roshbiz could regulate a theorem with saying literally that and subsequent formulations in terms of laws of requisite variety that the degrees of freedom on the inside have to correspond to the degrees of freedom on the things on the outside have to be controlled mainly the environment.

So I think that's absolutely right.

If you have already established through having an internal model of a particular behaviour, a particular way of deploying my actuators and making sense of that and, you know, being able to infer it was me that did that and this is me doing that and this is the kind of thing that I do,

that you are in a much stronger position to learn or incorporate or to imitate something that you have witnessed to novo.

And mathematically, the reason why that becomes easier is that the cost of belief updating in its most general mathematical sense.

So I'm not talking about personal beliefs.

I'm just talking about any belief entailed by the structure, the connectivity, the plasticity or indeed, say, neural activity.

Any belief updating comes along with a cost and that cost is can be variously articulated in terms of a failure to generalise informationally.

Thermodynamically, it has a cost of I-Landauer's principle.

Thermodynamically, it's interesting that the cost is the relative entropy between your prior and posterior beliefs.

So this will be like a computational cost, a complexity cost.

What that means is to make these steps, be they gradual or be they big jumps, they can only be made if they're, to use one of your words,

if they are suitably bounded in terms of the complexity cost, which means the degree of belief updating,

literally the relative entropy or the K-R divergence between the implicit posterior and prior after the update is sufficiently small,

which means literally you're changing your mind in a sort of mathematical sense in the most efficient way possible.

So if you're already close to and have the right kind of generative model to do imitation of a de novo behaviour in virtue of the fact you've done it before, that means that the complexity cost of actually committing to that and revising and belief updating your generative model

as now becomes within a tenable range, it becomes acceptable.

So that would certainly make sense if I was simulating that.

And we've been talking about ways of all sorts of different questions,

the questions in the chat, these questions and the plea from Michael,

how would you make this mechanics work in terms of resolving various hypotheses and evincing insights into how all of this works?

I think at the end of the day, you're going to have to simulate it.

And, you know, there's this a couple of this has not been done very much in the setting of the free energy principle or active inference, certainly.

But there now are a number of papers coming out within the past year or few months.

And, you know, it might be interesting to go to those.

And I'm thinking of, for example, a paper by Caspar Hesp's group that looked,

I think the final write-up is something like One Small Step for Mankind or something,

but it was originally entitled Ideas Worth Spreading.

So it's explicitly using the Bayesian mechanics of belief updating to look at how spreadable, how copyable certain beliefs were,

framed in the current age of the Internet and sort of tweets and the like.

And the same kind of theme has also been taken up by colleagues of Connor Hines and Maxwell Ranstead in terms of simulating epistemic communities,

where, coming back to Michael's use of the importance of epistemic spread and copying and imitation being basically an epistemic spread,

that's now starting to be simulated.

So I think we are on the cusp in the world of active inference, at least, on having mechanics out there that's actually realized in silica,

that you can start to intervene on a model things of this kind or that kind.

And speaking to Dean's question earlier on, you know, it strikes me that people in climatology must face exactly the same problem.

Is this missing data or did this thing actually go away?

And of course, it's a really important, really important question to answer.

Then in principle, you could do it if you've got a model of those legacy data.

But you have to have a model, which means you've got to basically have an in silico sort of digital twin of the way you think it works.

And then you can compare two gerative models or digital twins on the way it works and see which has the greatest evidence in light of the historical data.

That clearly hasn't been done either in climate change or your field.

But in principle, that would be the way to do that.

I'm sorry, I've gone over time.

Sorry.

It's totally fine.

My reflection on that as we again, anyone else who wants to have any last words is in the Par et al 2022 textbook, there's attention to two kinds of surprise.

It's plain surprise, which is an information theoretic quality, quantity associated with an observation with respect to its prior.

And the second being Bayesian surprise, which is what you just described, the cost of movement, which is related to the divergence of movement.

And I think in that difference between that was surprising and I surprised me or I have changed are two aspects of the scenario.

Because to just be surprised doesn't require one to change at all.

One can just say every day, wake up, check the news.

I can't believe that they're doing that.

You can just be surprised again and again.

And then there's the Bayesian surprise with the belief updating.

And that is really where the semantic advances of the Bayesian mechanics are just beginning to mature.

I hate the last word, but I hope it isn't.

So the propagations have been dropped in 53.0 and 53.1.

And Daniel and Dean have avoided being provocateurs so far.

What I'm really looking forward to is in 53.2, whether we can declare game on.

Because I think if we do, if we have that to look forward to, we can push the narrative to the end of the timetable.

And we can start to or continue to enhance the conversation that we initiated today.

So I just want to say thank you to all of you for all of the care and passion that you shared with us.

And I'm just happy as can be.

So thank you.

Thank you.

Any final comments, Hector or Michael?

No, just thank you for having us and sharing this time with us.

I mean, it was a great discussion.

Thank you.

Well, thank you very much.

It's been the most profitable session.

Excellent.

Carl, any last thoughts?

Again, I've enjoyed it greatly.

And, you know, it's a real pleasure to actually speak to all of you.

But in particular, Michael and Hector, you're actually using words.

So that's spoken words.

Very embodied.

Very inactive.

Lovely.

Excellent.

All right.

Till next time.

Thank you, everybody.

Bye.

Bye.

Thank you.

Thank you.

Thank you.